A description of how to run your script:

To run the script, you can use the following command structure:

python3 script_name.py --vcffile "your_vcf_file.vcf" --gfffile "your_gff_file.gff" --fastafile "your_fasta_file.fasta"

Replace script_name.py with the name of your Python script, and provide the required files for each argument. The --vcffile argument expects a VCF file (i.e., "assessmentData.vcf"), the --gfffile argument expects a GFF file (i.e., "PlasmoDB-54_Pfalciparum3D7.gff"), and the --fastafile argument expects a FASTA file (i.e., "PlasmoDB-54_Pfalciparum3D7_Genome.fasta"). All three arguments are compulsory, and you can replace the file names in quotes with your own file names. Make sure all files are in the same directory or give the correct file paths for the script to work.

A description of the input files required and their formats:

The script needs three input files: a GFF file **(**PlasmoDB-54_Pfalciparum3D7.gff), a VCF file (assessmentData.vcf), and a FASTA file (PlasmoDB-54_Pfalciparum3D7_Genome.fasta).
The GFF file (General Feature Format) contains gene annotations with columns like sequence ID (e.g., the chromosome where the feature is located), source, feature type (e.g., protein_coding_gene, mRNA, CDS), start and end positions which define the genomic coordinates of the feature, score (confidence or significance, often represented as . if unavailable), strand((forward + or reverse - strand), and attributes such as gene ID or description. The VCF file (Variant Call Format) stores genetic variations with columns such as chromosome(the reference sequence where the variant occurs), position (the 1-based coordinate of the variant), reference allele (REF) (the base present in the reference genome), alternate allele (ALT) (the variant base observed in samples), quality score (a measure of the confidence in the variant call), filter (whether the variant passes quality filters) and info (additional annotations like allele frequency, depth of coverage, or functional effects).
The FASTA file contains sequence data, where each entry starts with a > followed by an identifier, and the subsequent lines represent the DNA sequence.

A brief written description of how your script works:

This script analyses Single Nucleotide Polymorphism (SNP) data using a VCF file (for SNP details), a GFF file (for gene annotations), and a FASTA file (for DNA sequences). It processes SNPs with a quality score above 20, identifies whether the SNP is located in coding or non-coding regions, and checks if coding-region SNPs are synonymous (no change to the protein) or non-synonymous (causes a change to the protein).

The script uses GFF data to map SNPs to genome and derive the nucleotide and protein coordinates of the SNP if it is present in a coding region. Next, it mutates the DNA sequence by the ALT column from the VCF file to determine the impact on protein sequences. It produces a tab-separated output table summarising each SNP and its classification, logs the processing steps and records of SNPs that failed the quality checks and errors the script catches, and creates a bar plot showing the proportion of non-coding, synonymous, and

non-synonymous SNPs. The output is written to three files: a TSV table, a log file, and a bar plot image.

A description of the output files and an explanation of their contents:

The script generates three output files. The log file records the input filenames provided, counts the number of variants with QUAL ≤ 20 (failing quality checks), and confirms the location of the output files if the script runs successfully. If errors occur, such as missing files or invalid coordinates, they are reported in the log file for troubleshooting.

The bar plot visually represents the classification of variants with QUAL > 20, showing the proportions of non-coding variants (outside coding regions), synonymous variants (in coding regions but with no amino acid change), and non-synonymous variants (in coding regions causing an amino acid change).

The tab-separated table provides detailed information for each variant with QUAL > 20. It includes the chromosome, position, reference allele, alternate allele, variant type (non-coding, synonymous, or non-synonymous), transcript ID, protein location, reference amino acid, and alternate amino acid. Non-coding variants have 'NA' under transcript ID, protein location, reference amino acid, and alternate amino acid. For synonymous variants, the alternate amino acid column also contains 'NA' as there is no change to the protein sequence.

A description of any problems you've identified in the script with suggestions of how these could be overcome:

The script works well to generate the outputs asked in this assessment, but can be made more efficient and maintainable with a few improvements. The script has error handling. But we can enhance error handling by validating inputs early and adding specific exceptions for common issues. Improve the use of logging by systematically using levels like INFO, WARNING, and ERROR to distinguish normal updates from problems. The functions made in the script were long. We can try breaking some parts of the code into smaller pieces by using classes or more functions. These enhancements can make the code easier to debug.